

SinhalaJournal-LLM: A Style-Controlled Large Language Model for Diverse Sri Lankan Newspaper Writing

R26-SE-037

Project Proposal Report

B.Sc. (Hons) Degree in Information Technology Specialized in
Software Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The rapid expansion of digital journalism in Sri Lanka has increased the demand for fast, consistent, and engaging Sinhala news content. However, Sinhala-language media platforms continue to rely on manual headline writing and stock-image selection, leading to stylistic inconsistency and limited semantic alignment between textual and visual elements. Although transformer-based architectures have significantly advanced language modeling [1], multilingual large language models (LLMs) still exhibit performance disparities in low-resource languages such as Sinhala [2], [3], [4]. Existing Sinhala NLP research has primarily focused on text classification [5], continual language adaptation [6], and extractive summarization techniques [5], without addressing structured, style-controlled generative headline production.

This research proposes a Sinhala Style-Adaptive Headline and Image Generator that enables context-aware headline generation with explicit stylistic conditioning and semantically aligned visual prompts. Rather than training a foundation model from scratch, the study adopts a continual pretraining and fine-tuning strategy on an existing multilingual LLM, building upon Sinhala-adapted architectures such as SinLlama [6]. The proposed system integrates tokenizer adaptation, domain-specific fine-tuning using Sinhala news corpora, and controlled decoding mechanisms to generate headlines in multiple journalistic styles. The generated visual content is produced through integration with the Google Gemini Nano Banana API, guided by structured semantic prompts derived from generated headlines. The design also considers insights from digital journalism research highlighting the influence of headline framing on reader engagement and interpretation [7], [8].

System performance will be evaluated using quantitative metrics such as ROUGE and BLEU for content relevance, supported by human evaluation for stylistic appropriateness and journalistic accuracy. The expected outcome is an AI-driven framework that improves headline coherence, stylistic flexibility, and newsroom efficiency while supporting responsible media communication in low-resource language contexts.

Keywords: Sinhala NLP, Style-Controlled Headline Generation, Multimodal AI, Continual Pretraining, Digital Journalism.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor Prof. Nuwan Kodagoda and co-supervisor Ms. Poojani Gunathilake for their continuous guidance and support throughout this study. Their encouragement, constructive feedback, and insightful discussions greatly contributed to the development of this research.

I am also thankful to my project team members for their collaboration and shared efforts in planning and structuring the SinhalaJournal-LLM platform. Finally, I extend my appreciation to everyone who supported me with their valuable time and assistance during the completion of this proposal.

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LIST OF ABBREVIATIONS

API – Application Programming Interface
ACL – Association for Computational Linguistics
AACL-IJCNLP – Asia-Pacific Chapter of the Association for Computational Linguistics and International Joint Conference on Natural Language Processing
AI – Artificial Intelligence
BLEU – Bilingual Evaluation Understudy
BERT – Bidirectional Encoder Representations from Transformers
CMS – Content Management System
DSR – Design Science Research
GPU – Graphics Processing Unit
IDE – Integrated Development Environment
LLM – Large Language Model
LREC – Language Resources and Evaluation Conference
MADLAD – Multilingual and Document-Level Large Audited Dataset
MEGA – Multilingual Evaluation of Generative AI
NLP – Natural Language Processing
NER – Named Entity Recognition
PEFT – Parameter-Efficient Fine-Tuning
ROUGE – Recall-Oriented Understudy for Gisting Evaluation
SaaS – Software as a Service
SDG – Sustainable Development Goal
TF-IDF – Term Frequency–Inverse Document Frequency
UI – User Interface
UX – User Experience

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1. INTRODUCTION

1.1 Background and Context

The rapid digital transformation of media has significantly changed how news is created, distributed, and consumed. In Sri Lanka, online news platforms have become primary information sources, particularly among mobile-first audiences. Despite this shift, Sinhala-language journalism still relies heavily on manual headline writing and conventional image selection practices. This often results in stylistic inconsistency, slower publishing workflows, and limited semantic alignment between textual headlines and accompanying visuals.

Sinhala is classified as a low-resource language in global Natural Language Processing (NLP) research [2], [3]. Studies show that most large-scale NLP systems and benchmark datasets prioritize high-resource languages, leading to measurable performance disparities in multilingual models [2], [4]. Although transformer-based architectures have significantly advanced language modeling since the introduction of attention mechanisms [1], and models such as RoBERTa [9] and LLaMA 3 [10] have achieved strong generative performance, their reliability decreases in underrepresented languages. Multilingual evaluation frameworks further confirm instability and reduced consistency for low-resource languages in generative tasks [4], [11].

Efforts to address this imbalance include large multilingual corpora such as CulturaX [12] and MADLAD-400 [13], which expand language representation during pretraining. However, broader data inclusion does not automatically ensure domain-specific adaptability or stylistic control. Representation alone does not guarantee contextual or cultural alignment, particularly in specialized domains such as journalism.

Within the Sinhala NLP domain, several advancements have emerged. Dhananjaya et al. [5] demonstrated that multilingual transformer models can achieve strong performance in Sinhala text classification when properly fine-tuned. Aravinda et al. [6] further introduced SinLlama, a Sinhala-adapted decoder-based large language model developed through tokenizer enhancement and continual pretraining. This approach improved Sinhala language modeling without training a foundation model from scratch. Additionally, Sinhala text summarization research has explored extractive methods such as TF-IDF and TextRank for compressing online articles [14]. While these contributions improve representation and summarization, they do not address structured, style-controlled headline generation.

Headline generation differs from summarization or classification. It requires brevity, framing precision, and stylistic adaptation tailored to audience engagement. Research in digital journalism shows that headlines strongly influence reader perception and click behavior [7], and poorly framed headlines may contribute to persistent misinformation effects [8]. Therefore, automated headline generation systems must incorporate contextual accuracy and controlled stylistic variation.

Modern digital journalism is also increasingly multimodal, combining textual and visual elements to shape audience interpretation. However, Sinhala news platforms typically select images manually without semantic alignment mechanisms. Although global AI systems are beginning to support multimodal generation, these capabilities have not been adapted specifically for Sinhala journalism contexts. To address this limitation, the proposed system integrates structured semantic prompt generation with the Google Gemini Nano Banana API to produce contextually aligned visual outputs for generated headlines.

Despite progress in multilingual modeling [1]– [13] and Sinhala language adaptation [5], [6], no unified framework currently integrates continual LLM adaptation, explicit stylistic conditioning, and semantically aligned visual generation tailored to Sinhala digital newsroom workflows. This gap highlights the need for a structured, domain-aware generative system that leverages continual pretraining and fine-tuning strategies rather than attempting full foundation-model training.

This research aligns with Sustainable Development Goal (SDG) 4 (Quality Education), SDG 9 (Industry, Innovation and Infrastructure), and SDG 16 (Peace, Justice and Strong Institutions). Academically, it falls within the Artificial Intelligence and Data Science research cluster, specifically in Natural Language Processing and Generative AI, focusing on style-controlled generative modeling for low-resource language journalism.

1.2 Literature Review

Although several Sinhala NLP tools and multilingual language models have been developed, most focus on individual tasks such as classification [5], continual language adaptation [6], or extractive summarization [14]. While large transformer models show strong generative capability [1], [10], studies confirm inconsistent performance in low-resource languages [2], [3], [4], [11]. Despite these advancements, no unified system integrates grammar support, style-controlled headline generation, and visual alignment for Sinhala newsroom workflows. Therefore, this review examines multilingual LLM research [1]– [13], Sinhala-specific models [5], [6], summarization methods [14], and journalism studies [7], [8] to identify both progress and limitations from a system perspective.

Recent advances in Natural Language Processing (NLP) have been largely driven by transformer-based architectures, beginning with the attention mechanism introduced by Vaswani et al. [1]. This development enabled large-scale pre-trained models such as RoBERTa [9] and LLaMA 3 [10], which significantly improved performance in language understanding and generation tasks. These models demonstrate strong results in high-resource languages due to large training corpora and optimized model scaling strategies.

However, research consistently highlights disparities in language representation. Joshi et al. [2] demonstrate that NLP research and benchmark datasets disproportionately focus on a small number of dominant languages. Ranathunga and De Silva [3] further analyze the structural inequality in multilingual modeling and confirm measurable performance gaps for low-resource languages. Empirical evaluation studies show that multilingual large language

models often produce unstable or degraded outputs in non-English contexts [4], [11]. These findings indicate that multilingual inclusion alone does not guarantee reliable generative performance.

To mitigate representation imbalance, large multilingual datasets such as CulturaX [12] and MADLAD-400 [13] have been introduced to expand linguistic diversity during model pretraining. While these datasets improve token-level exposure for underrepresented languages, they do not inherently solve domain-specific or stylistic adaptation challenges. Multilingual exposure improves coverage, but it does not automatically ensure contextual or professional-domain suitability.

Within the Sinhala NLP domain, several significant contributions exist. Dhananjaya et al. [5] conducted a comprehensive evaluation of pre-trained multilingual and monolingual models for Sinhala text classification. Their study demonstrated that transformer-based encoders can achieve strong performance when fine-tuned on Sinhala datasets. However, the focus remained on classification tasks rather than generative applications.

Aravinda et al. [6] introduced SinLlama, a Sinhala-focused decoder-based large language model developed through tokenizer extension and continual pretraining. This approach improved Sinhala performance while avoiding full training from scratch. Although SinLlama enhances generative capacity, its evaluation primarily targeted classification benchmarks rather than structured headline generation or stylistic control.

Research on Sinhala text summarization has mainly explored extractive methods. Jahan and Wijesekara [14] compared TF-IDF and TextRank algorithms for summarizing Sinhala online articles. Their findings showed moderate improvements in content compression accuracy. However, extractive summarization selects existing sentences and lacks the flexibility required for concise, stylistically framed headline generation.

Beyond NLP, journalism research emphasizes the importance of headline framing. Kuiken et al. [7] demonstrate that digital headlines significantly influence reader engagement and interpretation. Additionally, misinformation studies show that misleading or poorly framed headlines can have persistent cognitive effects even after corrections [8]. These findings highlight the importance of contextual accuracy and responsible framing in automated headline systems.

Although modern generative models such as RoBERTa [9] and LLaMA 3 [10] provide powerful text generation capabilities, they do not incorporate structured stylistic conditioning tailored to newsroom contexts. Multilingual evaluation research also shows instability in generative consistency for low-resource languages [4], [11]. Existing Sinhala-focused models [5], [6] improve linguistic representation but do not integrate explicit style control, domain-specific headline optimization, or multimodal visual alignment.

Overall, the literature shows strong progress in multilingual modeling and Sinhala language adaptation. However, current approaches remain focused on classification, general language

modeling, or extractive summarization. There is limited research addressing controlled, style-adaptive generative headline production combined with semantically aligned visual outputs in the context of Sinhala digital journalism.

1.3 Research Gap

Existing Sinhala NLP research improves classification [5], generative adaptation [6], and extractive summarization [5], while multilingual models provide strong foundations [1], [10] but show instability in low-resource settings [2], [4], [11]. However, these approaches address isolated tasks rather than integrated newsroom needs. None combine explicit style control, structured headline optimization, and multimodal alignment within a single Sinhala-focused framework. This gap between technical improvements and practical newsroom integration defines the core research problem.

The reviewed literature demonstrates substantial progress in multilingual Natural Language Processing and Sinhala language adaptation. Transformer-based architectures [1] have enabled large-scale generative systems such as RoBERTa [9] and LLaMA 3 [10], while multilingual datasets like CulturaX [12] and MADLAD-400 [13] have expanded language representation. However, research consistently shows that increased multilingual coverage does not ensure reliable or consistent performance for low-resource languages such as Sinhala [2], [3], [4], [11]. Thus, representation improvement alone does not translate into domain-specific generative quality.

Within the Sinhala context, existing research has primarily concentrated on classification and representation learning. SinBERT demonstrated that multilingual transformer models can achieve competitive results for Sinhala text classification when properly fine-tuned [5]. SinLlama further enhanced generative capabilities by extending tokenization and performing continual pretraining on Sinhala corpora [6]. These approaches significantly improve language modeling performance compared to baseline multilingual systems. However, their evaluation focused mainly on classification benchmarks and general language tasks rather than structured headline generation.

Similarly, Sinhala text summarization research has largely adopted extractive approaches such as TF-IDF and TextRank [14]. While effective for content compression, extractive methods are fundamentally different from generative headline creation. They select existing sentences rather than producing concise, impact-driven headlines with controlled stylistic variation.

When compared to the practical requirements of digital journalism, a clear gap becomes evident. Journalism studies show that headlines directly influence reader engagement and interpretation [7], and poorly framed headlines may contribute to persistent misinformation effects [8]. Despite this, existing Sinhala NLP systems do not incorporate explicit mechanisms for stylistic conditioning, framing control, or headline-length optimization tailored to newsroom standards. Even advanced multilingual LLMs [10] provide general generative capabilities but lack structured adaptation for Sinhala journalistic contexts.

Evaluation studies further confirm instability in multilingual generative performance for low-resource languages [4], [11].

Moreover, modern digital news production increasingly integrates textual and visual elements. Current Sinhala-focused research does not address multimodal alignment between generated headlines and semantically appropriate visual content. Existing systems improve language representation, but they do not extend this capability to controlled multimodal newsroom workflows.

Therefore, although prior research has strengthened Sinhala language modeling and multilingual representation, none of the existing approaches integrate continual LLM adaptation, explicit style control, headline optimization, and multimodal alignment within a unified framework tailored to Sinhala digital journalism. Current systems enhance linguistic capability, yet they stop short of delivering domain-specific, style-adaptive generative solutions.

The research gap can therefore be clearly stated as follows:

There is no AI-driven framework that enables style-conditioned Sinhala headline generation with semantically aligned visual outputs using a continual pretraining and fine-tuning approach tailored specifically for digital newsroom applications.

Addressing this gap requires extending existing Sinhala-adapted large language models beyond general language tasks toward structured, domain-aware, and stylistically controlled generative systems suitable for real-world media environments.

2. OBJECTIVE

2.1 Main Objective

To design, implement, and evaluate a Sinhala Style-Adaptive Headline and Image Generation system by applying continual pretraining and fine-tuning on an existing multilingual large language model, enabling controlled stylistic headline generation and semantically aligned visual prompt production for digital newsroom applications.

2.2 Specific Objectives

To achieve the main objective, the following specific objectives are defined:

1. To develop a structured Sinhala news dataset with style annotations

This objective focuses on constructing a high-quality Sinhala news corpus suitable for controlled headline generation. Full-length news articles and their original headlines will be collected from established Sinhala digital media platforms. The dataset will be cleaned, tokenized, and structured into article–headline pairs. In addition, headlines will be categorized into predefined journalistic styles such as formal reporting, breaking news, youth-oriented tone, and editorial commentary. These annotations will enable supervised fine-tuning and style-control evaluation. The dataset will be split into training, validation, and testing subsets to ensure experimental reliability.

2. To perform continual pretraining and domain-specific fine-tuning of a Sinhala-adapted multilingual LLM for headline generation

Rather than training a foundation model from scratch, this objective involves adapting an existing multilingual large language model (e.g., SinLlama) through continual pretraining on a Sinhala news corpus. Tokenizer enhancement and domain-specific fine-tuning will be applied to optimize the model for concise headline generation. The model will be trained to transform full news articles into short, semantically accurate headlines. Performance will be compared against baseline multilingual model outputs to measure improvements in relevance and coherence.

3. To implement a style-conditioning mechanism for controlled headline generation

This objective aims to integrate an explicit style-control mechanism into the generative process. A structured approach, such as prompt conditioning or style embeddings, will be designed to guide the model to generate headlines in a specified journalistic tone. The system will be tested to ensure that style variation does not compromise factual accuracy or semantic alignment with the source article. Style classification accuracy will be measured using an external classifier or validation model to verify control effectiveness.

4. To design a semantic visual prompt generation module aligned with generated headlines

This objective addresses the multimodal aspect of the system. Key entities, themes, and contextual signals will be extracted from generated headlines using named entity recognition and keyword extraction techniques. These elements will be transformed into structured visual prompts and transmitted to the Google Gemini Nano Banana API for image generation. The returned visual outputs will be validated through a multimodal alignment mechanism to ensure semantic consistency with the generated headline. The semantic consistency between headline and generated visual prompt will be evaluated through similarity scoring and human assessment to ensure contextual alignment.

5. To evaluate the proposed system using quantitative and qualitative metrics

This objective ensures systematic validation of the developed system. Quantitative evaluation will include ROUGE and BLEU scores to measure headline relevance and content overlap with reference headlines. Style control effectiveness will be measured using classification accuracy and confusion matrix analysis. Additionally, structured human evaluation will be conducted with journalism students or media practitioners to assess clarity, stylistic appropriateness, factual correctness, and newsroom usability. Comparative analysis will be performed against baseline multilingual LLM outputs to demonstrate measurable improvement.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology combined with Experimental System Evaluation, aligned with Software Engineering principles. The primary contribution is the design and implementation of a novel artefact — a Sinhala Style-Adaptive Headline and Image Generation system — followed by structured empirical validation.

The Design Science component focuses on constructing and refining the generative framework, including continual model adaptation, style-conditioning mechanisms, and semantic visual alignment. The Experimental component evaluates system performance using controlled baselines and quantitative metrics.

This study does not train a foundation model from scratch. Instead, it builds upon a Sinhala-adapted multilingual large language model, specifically SinLlama [6], and applies continual pretraining and domain-specific fine-tuning to achieve style-controlled headline generation. This approach ensures computational feasibility while maintaining research novelty.

3.2 Technical Methods and Algorithms

3.2.1 Continual Pretraining and Fine-Tuning Strategy

Transformer architectures, introduced through the attention mechanism [1], form the backbone of modern generative systems such as RoBERTa [9] and LLaMA 3 [10]. However, multilingual models often underperform in low-resource languages due to representation imbalance [2], [4].

To address this limitation, the proposed system adopts a continual pretraining strategy using SinLlama [4], which extends multilingual LLM capabilities for Sinhala. The technical procedure includes:

- Tokenizer enhancement to improve Sinhala lexical coverage
- Continued pretraining on a curated Sinhala news corpus
- Supervised fine-tuning for article-to-headline generation

This strategy improves domain adaptation without requiring full-scale foundational model training.

3.2.2 Headline Generation Model

Headline generation is formulated as a sequence-to-sequence transformation task:

Input: Full Sinhala news article

Output: Concise, context-aware headline

The fine-tuned SinLlama model generates headlines using conditional prompting and controlled decoding mechanisms. Decoding parameters such as temperature and top-k sampling will be adjusted to balance creativity and factual stability. Headline length constraints will be applied to maintain newsroom standards.

Performance will be compared against the unmodified multilingual LLaMA-based baseline [10] to demonstrate measurable improvement achieved through Sinhala-specific fine-tuning.

3.2.3 Style-Conditioning Mechanism

To enable explicit stylistic control, a conditioning mechanism will be integrated into the generation process. Style categories (formal, breaking news, youth-oriented, editorial) will be encoded using structured prompt tokens or style embeddings appended to input sequences.

Style effectiveness will be evaluated using a classification-based validation approach, where generated headlines are tested for alignment with their intended style labels. This ensures that stylistic variation is measurable and not implicitly generated.

3.2.4 Semantic Visual Prompt Generation

Modern digital journalism increasingly integrates textual and visual elements to enhance audience engagement. To support this multimodal requirement, the proposed system generates structured visual prompts derived from the generated headlines.

First, key entities, themes, and contextual signals are extracted from the generated headline using Named Entity Recognition (NER) and keyword extraction techniques. These extracted elements are transformed into structured semantic prompts designed to accurately represent the core meaning of the headline.

Rather than developing a custom image generation model, the system integrates with the **Google Gemini Nano Banana API** for visual rendering. The structured semantic prompts are transmitted to the external API, which returns a generated image corresponding to the provided textual description.

To ensure contextual reliability, a Multimodal Alignment Validation Layer evaluates the semantic consistency between the generated headline and the returned visual output. Semantic similarity scoring techniques and structured human evaluation are applied to assess whether the image accurately reflects the intended meaning of the headline.

This architecture ensures that the research contribution remains focused on semantic prompt engineering, alignment validation, and controlled multimodal integration rather than foundational image model development.

3.3 Tools and Platforms with Justification

The system is implemented using the following technologies:

Tool / Platform	Purpose	Justification
SinLlama	Base LLM	Sinhala-adapted decoder LLM
PyTorch	Model training	Industry-standard deep learning framework
HuggingFace Transformers	Fine-tuning & tokenizer handling	Efficient LLM experimentation
PEFT (LoRA)	Parameter-efficient adaptation	Parameter-efficient adaptation that reduces training cost.
Google Colab / GPU Server	Training environment	Required for LLM fine-tuning
ROUGE, BLEU	Evaluation metrics	Standard NLP evaluation
Google Gemini Nano Banana API	External image generation service	Enables structured semantic prompt-based visual rendering without custom model training

Table 1: Tools and Platforms with Justification

These tools are selected based on compatibility with transformer architectures and strong research adoption in NLP studies [2], [3].

3.4 Data Collection and Ethical Considerations

3.4.1 Dataset Preparation

A structured Sinhala news dataset will be constructed using publicly accessible digital news articles. Each instance will contain:

- Full news article
- Corresponding headline
- Annotated journalistic style label

The dataset will be divided into training, validation, and testing subsets to ensure experimental rigor.

Ethical considerations include:

- Use of publicly available content only
- No collection of personal or sensitive information
- Responsible use of generated outputs
- Informed consent for human evaluation participants

The study will strictly focus on academic evaluation and responsible AI usage.

3.5 Validation Strategy and Performance Metrics

System validation will include both quantitative and qualitative evaluation to ensure comprehensive assessment.

3.5.1 Quantitative Evaluation

1. **ROUGE Score** – Measures n-gram overlap between generated and reference headlines [14]
2. **BLEU Score** – Measures precision-based textual similarity
3. **Style Classification Accuracy** – Measures effectiveness of style-conditioning control
4. **Headline Length Compliance Rate** – Measures adherence to brevity constraints

Results will be compared against:

- Unmodified multilingual LLM baseline [10]
- Fine-tuned model without style conditioning

This ensures measurable performance improvement beyond general multilingual modeling [4], [11].

3.5.2 Qualitative Evaluation

Human evaluation will assess:

- Clarity and readability
- Stylistic appropriateness
- Factual consistency
- Newsroom usability

Participants (e.g., journalism students or media practitioners) will rate outputs using structured evaluation criteria, ensuring real-world relevance.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The SinhalaJournal-LLM system is designed as a modular AI-driven newsroom assistance platform that supports multiple Sinhala newspaper writing tasks. The overall system consists of four primary components:

1. Sinhala Grammar Checker
2. Style-Adaptive Headline and Image Generator (Individual Component)
3. Sinhala Style Rewriter
4. Sinhala News Summarizer

Each component operates independently but shares a common underlying Sinhala-adapted Large Language Model (SinLlama) [6], which is fine-tuned using domain-specific Sinhala news data. The system follows a modular architecture to ensure scalability, maintainability, and clear separation of responsibilities.

The core research contribution of this individual project lies within the **Style-Adaptive Headline and Image Generator**, which introduces controlled headline generation and semantic visual alignment. For visual rendering, the system integrates with the Google Gemini Nano Banana API as an external AI service. The Image Generation Module does not internally generate images; instead, it converts semantic headline representations into structured visual prompts that are transmitted securely to the external API. The returned visual output is then processed through a Multimodal Alignment Layer before being delivered to the user.

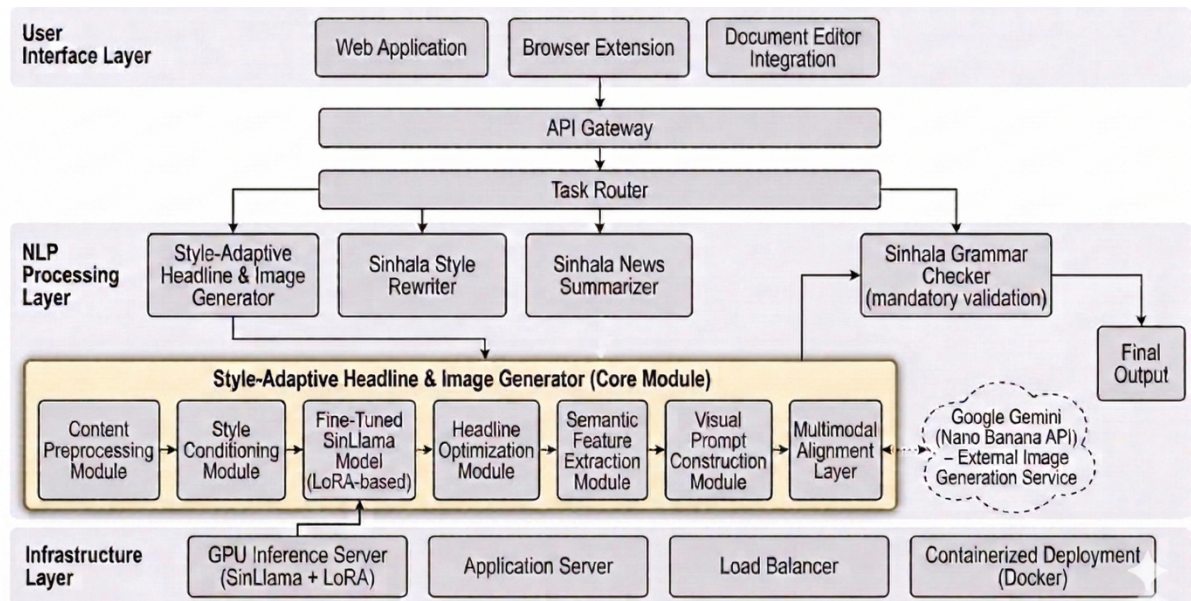


Figure 1: High level system architecture

Figure 4.1 illustrates the layered architecture of the proposed SinhalaJournal-LLM system, with specific emphasis on the Sinhala Style-Adaptive Headline and Image Generator component. The architecture follows a modular and layered design to ensure maintainability, scalability, and clear separation of responsibilities.

4.2 Component Interactions

As shown in Figure 4.1, the SinhalaJournal-LLM system follows a layered interaction model where each component communicates through clearly defined interfaces. When a user submits a Sinhala news article, the request moves from the User Interface Layer to the Application Layer, where authentication, validation, and routing are handled before passing it to the NLP Processing Layer.

Within the highlighted Sinhala Style-Adaptive Headline and Image Generator, the Content Preprocessor prepares the article through cleaning and tokenization. The Style Controller then injects the selected tone before sending the structured input to the fine-tuned SinLlama model [6]. As illustrated in Figure 4.1, the Headline Generation Engine works together with the Headline Optimization Module to ensure the output meets length and clarity constraints.

After generating the headline, semantic features are sent to the Image Generation Module. This module converts extracted entities and contextual information into structured visual prompts, which are transmitted to the Google Gemini Nano Banana API as an external image generation service. The returned image is then passed through the Multimodal Alignment Layer to verify semantic consistency between the generated headline and visual output before producing the final result. At the system level, components communicate through modular APIs, allowing shared access to the fine-tuned model while maintaining separation between tasks such as grammar checking and summarization.

4.3 Data Flow Explanation

As depicted in Figure 4.1, the data flow follows a structured pipeline from input to final output.

The process begins with a full Sinhala news article and a selected style label. The article first goes through preprocessing, including tokenization and normalization. The processed content, combined with the style parameter, is then passed to the fine-tuned SinLlama model [6].

The model generates a headline using controlled decoding settings. This headline is refined by the Headline Optimization Module to ensure it meets newsroom standards such as brevity and emphasis. The finalized headline is then used to extract key entities and themes.

These extracted elements are converted into structured semantic prompts, which are transmitted to the Google Gemini Nano Banana API for visual rendering. The generated image is then validated by the Multimodal Alignment Layer to ensure it matches the headline meaning. The generated image is validated by the Multimodal Alignment Layer to ensure it matches the headline meaning. Finally, the system returns both the headline and image to the user. As shown in Figure 4.1, this flow keeps each stage clearly separated while maintaining overall system integration.

4.4 Scalability and Security Considerations

As illustrated in Figure 4.1, the architecture is designed to support scalability and security.

For scalability, the model-serving layer is separated from the application layer, allowing GPU-based inference to scale independently as demand increases. The fine-tuned SinLlama model [6] can be deployed through APIs, enabling horizontal scaling with multiple inference instances. The modular design also makes it easier to add future components without restructuring the entire system. Since image rendering is handled by an external API service, computational overhead on the internal GPU inference server is reduced, allowing better scalability for headline generation tasks.

For security, input validation is handled at the Application Layer to prevent invalid or harmful requests. Rate limiting and authentication control access to the system. Since only publicly available news content is processed, sensitive data storage is minimized. Secure storage and controlled access are applied to training datasets and model checkpoints. Secure API key management is implemented for the Google Gemini Nano Banana API integration. Authentication credentials are stored using environment-level configuration and are not exposed within the application source code.

Additionally, the multimodal validation layer helps reduce the risk of producing misleading or inconsistent outputs. These measures ensure the system remains reliable, secure, and suitable for real-world deployment.

5. USER REQUIREMENTS

5.1 Stakeholder identification

The primary stakeholders of the Sinhala Style-Adaptive Headline and Image Generator include journalists and news editors who use the system to generate style-controlled headlines and supporting visuals for digital publication. Digital media organizations benefit from improved productivity and consistency in newsroom workflows. Journalism students and educators may use the system as a learning support tool for understanding headline framing and stylistic variation. System administrators are responsible for deployment, monitoring, and maintaining the infrastructure, while developers and researchers maintain the fine-tuned SinLlama-based model and improve system performance. Each stakeholder interacts with the system at different levels, from direct content generation to technical maintenance and evaluation.

5.2 Functional requirements

Article Input Handling

The system shall accept a complete Sinhala news article through a web interface or API endpoint. The input must support standard Sinhala Unicode text and allow sufficient length for full news content. The system shall validate the input format before processing to ensure that the content is suitable for headline generation.

Style Selection:

The system shall provide users with predefined headline style options such as formal, breaking news, youth-oriented, editorial, and sports. The selected style shall be explicitly injected into the generation process through prompt conditioning or style embedding. The system shall ensure that the selected style influences tone and framing without altering factual meaning.

Text Preprocessing:

The system shall preprocess the submitted article by performing tokenization, normalization, and noise removal. This includes handling punctuation, special characters, and formatting inconsistencies. The preprocessing module shall prepare structured input compatible with the fine-tuned SinLlama model.

Style-Controlled Headline Generation:

The system shall generate a concise and context-aware headline using the continually pre-trained and fine-tuned SinLlama model. The model shall transform the full article into a short headline while preserving key information. Controlled decoding parameters (e.g., temperature, top-k sampling) shall be applied to balance creativity and factual stability.

Headline Optimization:

After initial generation, the system shall refine the headline to meet newsroom standards.

This includes enforcing maximum length constraints, improving clarity, and ensuring keyword emphasis. The system shall verify that the final headline remains grammatically correct and semantically aligned with the source article.

Semantic Image Prompt Generation:

The system shall extract key entities, themes, and important contextual elements from the generated headline using semantic analysis techniques. These elements shall be converted into a structured visual prompt suitable for AI-based image generation tools.

Image Generation and Alignment:

The system shall generate a context-relevant image by transmitting structured visual prompts to an external image generation service (Google Gemini Nano Banana API). The returned image shall be validated through a multimodal alignment mechanism to ensure semantic consistency with the generated headline. If inconsistency is detected, the system shall regenerate or refine the image prompt before returning the final output.

External API Communication:

The system shall securely transmit structured visual prompts to the Google Gemini Nano Banana API and handle API responses, timeouts, and error conditions appropriately.

Output Delivery:

The system shall return the final style-controlled headline and associated visual output in a clearly formatted structure. Users shall be able to review and edit the generated content before publication. The system shall also log generation details for monitoring and evaluation purposes.

5.3 Non-functional requirements

Performance Requirements

1. The system shall generate a style-controlled headline within an average response time of less than 5 seconds under normal operating conditions.
2. The system shall generate both headline and associated image within an average response time of less than 10 seconds when GPU inference is available.
3. The system shall achieve a minimum ROUGE-L score of 0.75 when compared with reference headlines during evaluation.

Security Requirements

1. The system shall validate all user inputs to prevent prompt injection and malformed requests.
2. The system shall restrict access to administrative functions through authentication and authorization mechanisms.
3. The system shall securely store training datasets and model checkpoints using controlled access permissions.

4. The system shall protect API keys and authentication credentials used for external image generation services through secure environment configuration and access control mechanisms.

Usability Requirements

1. The system shall provide a simple and intuitive interface requiring minimal technical knowledge.
2. The system shall allow users to select headline style through a clearly visible control panel.
3. The system shall allow users to review and edit generated headlines before publication.

Scalability Requirements

1. The system shall support horizontal scaling of model-serving infrastructure through containerized or cloud-based GPU deployment.
2. The system shall maintain stable performance when handling multiple simultaneous user requests.

Reliability Requirements

1. The system shall maintain consistent output quality across repeated requests for similar inputs.
2. The system shall provide meaningful error messages if generation fails or if input is invalid.
3. The system shall handle external API failures gracefully by providing meaningful error messages or fallback mechanisms without affecting core headline generation functionality.

Maintainability Requirements

1. The system shall allow model updates and fine-tuning without requiring major architectural redesign.
2. The system shall support extension to additional headline styles or new NLP components through modular integration.

6. COMMERCIALIZATION PLAN

6.1 Target market and customer persona

The primary target market for SinhalaJournal-LLM consists of Sri Lankan media organizations, digital news platforms, and publishing houses that produce Sinhala content daily. These organizations require fast, accurate grammar validation, headline generation, summarization, and style consistency under tight deadlines.

Secondary markets include universities, journalism institutes, and Sinhala language departments that require structured writing support tools. Individual users such as bloggers, freelance journalists, and content creators also represent a growing user base.

A typical customer persona is a newsroom editor who reviews multiple articles per day and needs automated assistance to improve accuracy, maintain tone, and generate publish-ready content efficiently.

6.2 Value proposition

SinhalaJournal-LLM provides a unified AI platform integrating:

- Sinhala Grammar Checker
- Sinhala Headline Generator
- Sinhala Style Rewriter
- Sinhala News Summarizer

Unlike global AI tools that lack Sinhala support, this system is specifically optimized for Sinhala journalism. It improves editorial efficiency, reduces proofreading time, ensures stylistic consistency, and accelerates digital publishing workflows.

The platform offers an end-to-end solution tailored to a low-resource language context. The system also integrates AI-based image generation using an external multimodal API, enabling automatic headline–image alignment for digital publishing workflows.

6.3 Revenue model

The system follows a Software-as-a-Service (SaaS) model with recurring subscription revenue.

1. Institutional Subscription Plans

Media organizations can subscribe under tiered plans:

- **Basic Plan** – Limited monthly corrections and summarizations
- **Professional Plan** – Higher request limits with priority processing
- **Enterprise Plan** – Unlimited usage, API access, analytics, and dedicated support

Annual subscription packages may be offered at discounted rates.

2. API-Based Pricing

Third-party news platforms, content management systems, and mobile applications can integrate SinhalaJournal-LLM via APIs. Pricing may follow:

- Per-request billing
- Per-token usage pricing
- Custom enterprise API agreements

3. Educational Licensing

Universities and institutes may receive discounted academic licenses to support journalism education and research.

4. Freemium Model

Individual users may access limited monthly usage for free, with optional premium upgrades for advanced features.

6.4 Cost estimation

The main operational costs include:

- GPU cloud hosting for SinLlama inference
- External multimodal API usage costs (Google Gemini Nano Banana API)
- API servers and load balancing
- Data storage and monitoring systems
- System maintenance and updates

GPU infrastructure represents the highest recurring cost. However, the use of parameter-efficient fine-tuning techniques (such as LoRA) reduces retraining and computational expenses compared to full model training.

Operational sustainability can be achieved if several mid-scale media organizations subscribe to the Professional Plan. For example, 10 organizations subscribing at mid-tier pricing can cover infrastructure and maintenance costs. Since image generation is handled through an external API, operational costs include per-request or usage-based billing. Therefore, pricing and subscription plans must account for variable API consumption depending on the number of generated images.

6.5 Pricing strategy

Pricing must balance affordability within the Sri Lankan market and long-term sustainability. A possible structure may include:

- **Basic Plan** – LKR 8,000 per month
- **Professional Plan** – LKR 25,000 per month
- **Enterprise Plan** – Custom pricing

Educational institutions may receive discounted pricing. API usage may follow per-request or per-token billing to ensure fairness based on consumption. Image generation requests may be counted separately within subscription tiers due to external API costs.

The pricing strategy reflects the cost savings compared to hiring additional editorial staff while maintaining competitive affordability.

6.6 Competitive advantage

SinhalaJournal-LLM offers:

- Dedicated support for Sinhala language
- Journalism-focused optimization

- Hybrid rule-based and LLM-based architecture
- Integration of multiple writing tools in one platform

Most existing solutions either lack Sinhala support or provide limited spell-checking features. This system fills a significant technological gap in the Sri Lankan media industry.

6.7 Intellectual property considerations

The intellectual property of the system includes:

- Fine-tuned Sinhala LLM model weights
- Hybrid grammar correction and style adaptation algorithms
- Journalism-specific datasets
- API integration and orchestration framework

The base large language model remains subject to its original licensing terms. Image generation is performed via Google Gemini Nano Banana API, and generated outputs are subject to the terms and conditions of the external service provider. Fine-tuned weights, integration logic, and custom-developed modules may be protected under institutional intellectual property policies.

All deployments will comply with open-source licensing requirements and ethical AI usage standards.

7. BUDGET AND JUSTIFICATION

The estimated budget for the SinhalaJournal-LLM project reflects realistic infrastructure requirements for model fine-tuning, experimentation, multi-component deployment, and evaluation.

Since the system integrates multiple NLP components powered by a fine-tuned Sinhala Large Language Model (SinLlama) using Parameter-Efficient Fine-Tuning (LoRA), GPU-based computation is required during experimentation and training phases.

The budget also considers API hosting, system integration, testing, and demonstration deployment to represent the true economic scale of a production-ready AI system.

7.1 Hardware costs

Item	Estimated Cost (LKR)	Justification
Cloud GPU Rental (NVIDIA T4 / A100 equivalent)	120,000	Approx. 20 USD/hour × 40 hours fine-tuning & experimentation
Development Laptop	0	Existing personal device

Table 2: Hardware Cost Estimation

Estimated Hardware Cost: 100,000 – 120,000 LKR

Justification:

Fine-tuning transformer-based models for multiple tasks (grammar correction, headline generation, style rewriting, summarization) requires GPU acceleration. Instead of purchasing a dedicated GPU (which would exceed 300,000 LKR), cloud GPU rental provides a cost-effective and scalable solution. Using Parameter-Efficient Fine-Tuning (LoRA) significantly reduces memory requirements and training time compared to full model retraining.

7.2 Software licenses

Software	Cost (LKR)	Justification
Python	0	Open-source
PyTorch	0	Open-source
HuggingFace Transformers	0	Open-source
PEFT (LoRA)	0	Open-source
Docker	0	Open-source
VS Code	0	Open-source

Table 3: Software License Cost Estimation

Estimated Software Cost: 0 LKR

Justification:

The system relies entirely on open-source technologies, eliminating licensing costs.

7.3 Cloud services

Assumptions:

- 2 vCPU, 4GB RAM instance
- ~10,000 LKR per month
- 6 months usage

Service	Monthly Cost (LKR)	Duration	Total Cost (LKR)
Cloud VM Hosting	10,000	6 months	60,000
Domain Registration	5,000	One-time	5,000
AI Image Generation API (Gemini)	0	Research usage	0

Table 4: Cloud Service Cost Estimation

Estimated Cloud Cost: 65,000 LKR

Justification:

The cloud infrastructure hosts:

- REST API gateway
- Grammar validation module
- Headline generation module
- Style rewriting module
- Summarization module
- Demo web application
- Monitoring and logging services

Full LLM training does not run continuously. GPU usage is limited to the fine-tuning phase, while inference services operate on optimized deployments. The image generation component integrates the Google Gemini API under its current research free-tier availability. Therefore, no direct API cost is incurred during the academic experimentation phase. However, future commercial deployment may require usage-based billing depending on API pricing policies.

7.4 Human effort cost

Although this is an academic project, the development effort can be estimated to demonstrate the project scale and economic value.

Activity	Estimated Hours
Dataset preparation & annotation	80 hrs
Model fine-tuning & experimentation	120 hrs
System development & API integration	100 hrs
Evaluation & testing	60 hrs
Total Estimated Effort	~360 hours

Table 5: Human Effort Cost Estimation

Cost Estimation

Assumptions:

- Intern rate = 1,500 LKR per day
- 1 working day = 8 hours

Total working days:

$360 \text{ hours} \div 8 \text{ hours/day} = 45 \text{ days}$

Estimated Human Effort Cost:

$45 \text{ days} \times 1,500 \text{ LKR/day} = \mathbf{67,500 \text{ LKR}}$

7.5 Total Estimated Budget

Category	Estimated Cost (LKR)
Hardware (GPU Rental – if required)	0 – 32,000
Software	0
Cloud Hosting & Deployment	38,400 – 43,400
Human Effort (Intern basis)	67,500
Total Estimated Direct Cost	105,900 – 142,900 LKR

Table 6: Total Cost Estimation

Overall Budget Justification

The proposed budget is intentionally minimized through:

- Use of institutional GPU infrastructure
- Adoption of open-source frameworks
- Parameter-Efficient Fine-Tuning (LoRA) to reduce computational overhead
- Limited-duration cloud hosting for deployment and demonstration

The primary financial cost arises from cloud hosting and development effort rather than expensive hardware acquisition.

This approach ensures financial feasibility while maintaining research quality, scalability potential, and technical robustness. The cost structure also demonstrates that SinhalaJournal-LLM is economically viable for both academic research and future commercialization.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Overview of Work Breakdown

The project is divided into the following major phases:

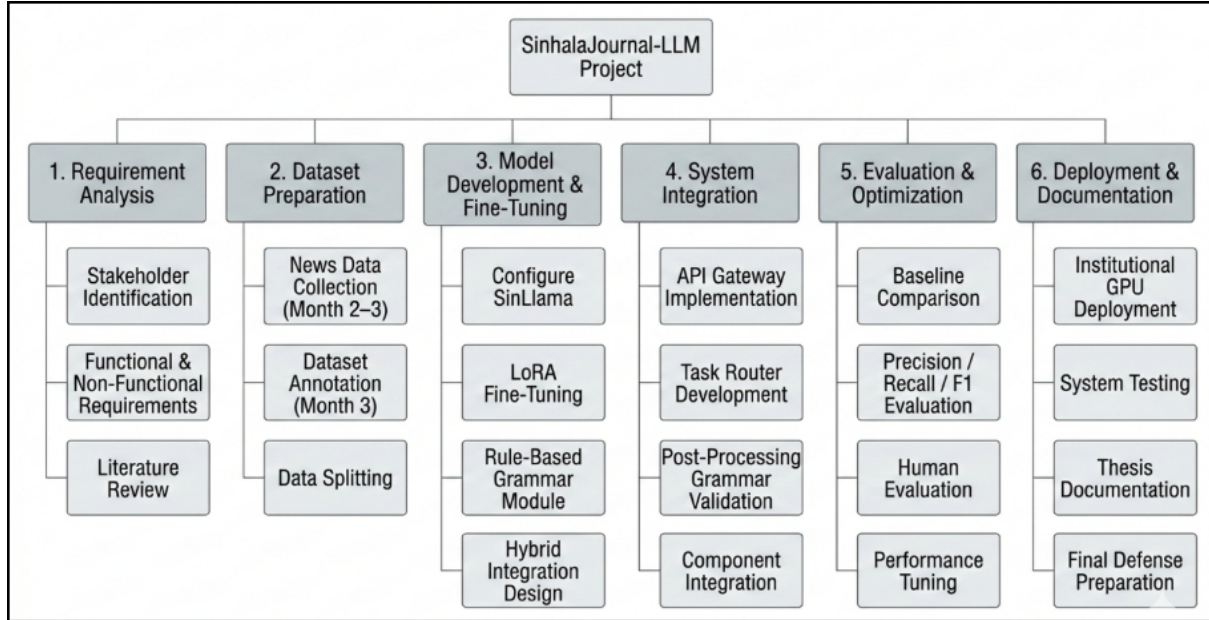


Figure 2: Work Breakdown Structure

Figure 2 presents the hierarchical Work Breakdown Structure (WBS) of the SinhalaJournal-LLM project. The project is organized into six major phases with clearly defined tasks to ensure structured development across the one-year timeline. The Sinhala Grammar Checker activities—LoRA fine-tuning, rule-based validation, and hybrid integration—are highlighted to reflect the individual research contribution. The WBS also demonstrates balanced workload distribution and coordinated system-level integration.

8.2 Timeline alignment

The project spans twelve months, following a phased and progressive development schedule:

- Months 1–2: Requirement analysis and literature review
- Months 2–3: Dataset collection
- Month 3: Dataset annotation
- Months 4–6: SinLlama fine-tuning and rule-based module development
- Months 6–7: Hybrid integration and system integration
- Months 7–9: Experimental evaluation and optimization
- Months 9–10: Institutional deployment and system testing
- Months 7–12: Documentation and thesis writing
- Months 10–12: Final validation and defense preparation

This timeline allows iterative experimentation, refinement, and integration while maintaining feasibility within the academic year.

8.3 Task Breakdown Per Member

Phase	Member 1 – Grammar Checker	Member 2 – Headline Generator	Member 3 – Style Rewriter	Member 4 – News Summarizer
Requirement Analysis	Define objectives and evaluation metrics	Define generation objectives and style criteria	Define transformation objectives and tone categories	Define summarization objectives and evaluation criteria
Dataset Preparation	Prepare annotated error–correction dataset	Prepare labeled headline dataset	Prepare style transformation dataset	Prepare summarization dataset
Model Development	Implement <u>LoRA</u> fine-tuning and rule-based validation	Implement style-conditioned fine-tuning	Implement instruction-based fine-tuning	Implement summarization model
System Integration	Develop API and integrate with router	Develop API and integrate with router	Develop API and integrate with router	Develop API and integrate with router
Evaluation	Evaluate using Precision, Recall, F1, GLEU	Evaluate relevance and style consistency	Evaluate tone preservation and consistency	Evaluate using ROUGE and baseline comparison

Figure 3: Task Breakdown Per Member

8.4 Balanced workload distribution

Workload distribution is structured to ensure fairness and technical parity among team members. While the Sinhala Grammar Checker involves both neural fine-tuning and rule-based development, other components require comparable effort in dataset preparation, model adaptation, evaluation, and integration.

Shared tasks such as API implementation, routing logic, and system testing further balance technical contribution. This structured allocation ensures that each member contributes meaningfully to both individual research objectives and overall platform development.

9. GANTT CHART

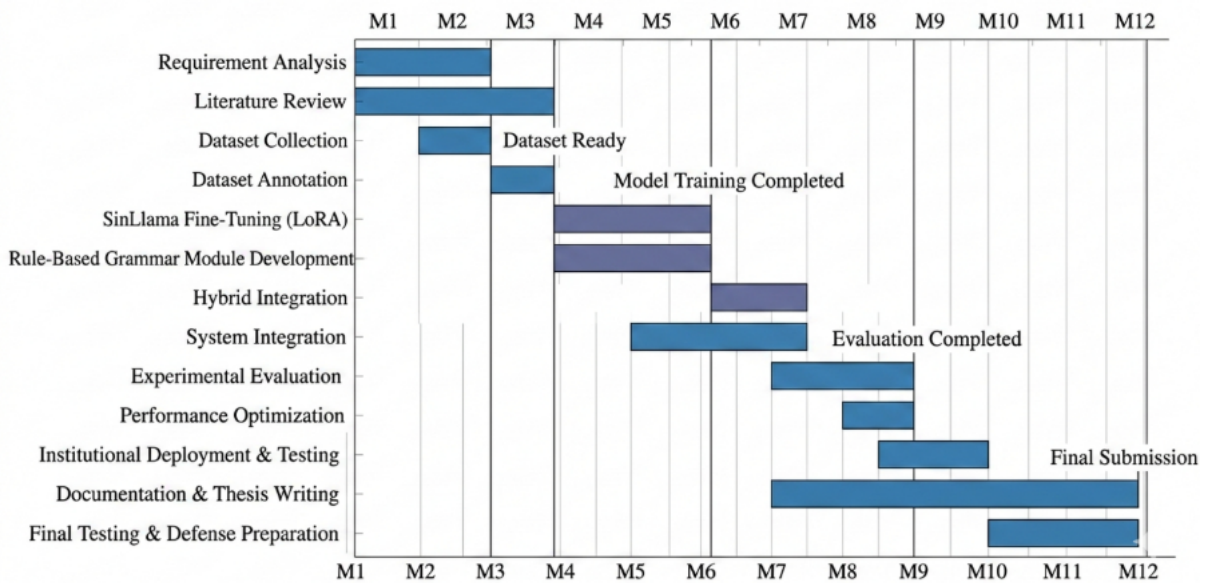


Figure 4: 12 Month project timeline Gantt chart

Figure 3 presents the 12-month project timeline in the form of a Gantt chart. The schedule aligns research, development, integration, and evaluation phases in a structured and progressive manner. Early months focus on requirement analysis and dataset preparation, followed by model fine-tuning and rule-based module development. System integration and experimental evaluation occur in the middle phase, while deployment, documentation, and final validation are scheduled in the concluding months.

Milestones are clearly defined to monitor progress, including dataset completion, model training completion, evaluation finalization, and final submission. The timeline ensures feasibility, iterative refinement, and sufficient preparation for project defense.

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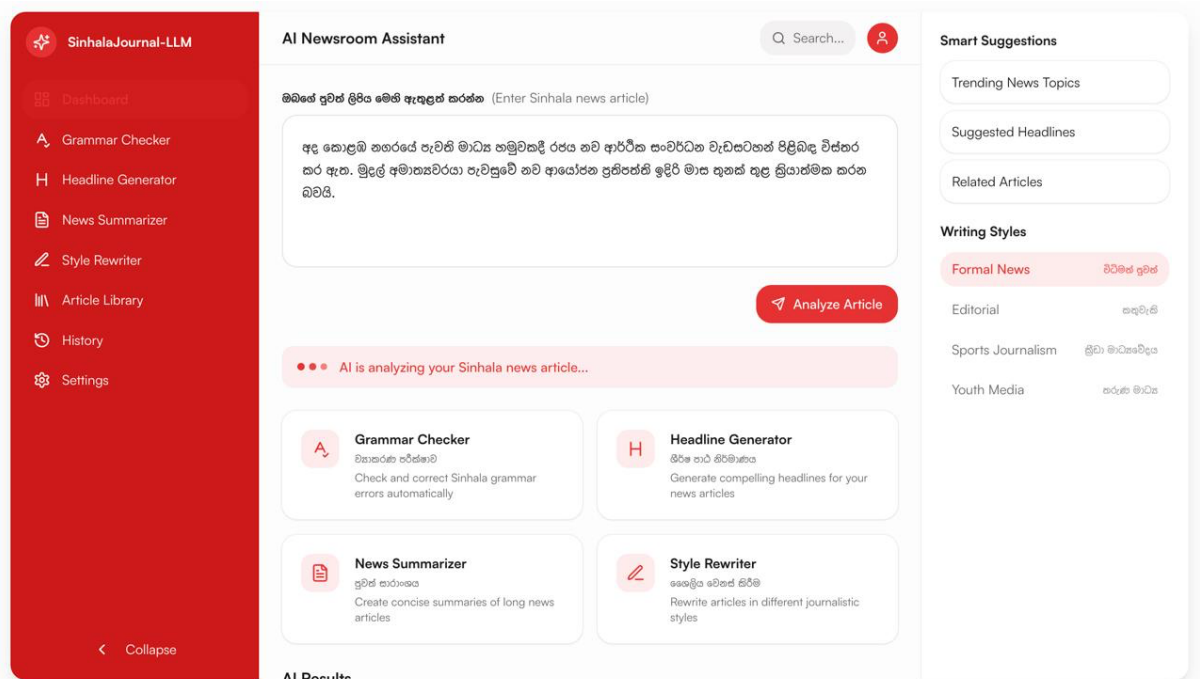
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11. APPENDICES

11.1 Preliminary UI mockups

Appendix 1: Preliminary UI mockups

- Web UI



- Web Extension UI



SinhalaJournal-LLM

AI Writing Assistant

DETECTED ARTICLE

අද කොළඹ නගරයේ පැවති මාධ්‍ය හමුවකදී රජය නව ආර්ථික සංවර්ධන වැඩසටහන් පිළිබඳ විස්තර කර ඇත...

QUICK AI ACTIONS



Check Grammar

ව්‍යාකරණ පරීක්ෂාව



Generate Headline

ශීර්ෂ පාඨ



Summarize

සාරාංශය



Rewrite Style

ගෙනලිය

... Analyzing article...



Corrected Text

අද කොළඹ නගරයේ පැවති මාධ්‍ය හමුවකදී, රජය නව ආර්ථික සංවර්ධන වැඩසටහන් පිළිබඳ විස්තර කර ඇත.

1 correction



Suggested Headlines

- > නව ආර්ථික ප්‍රතිපත්ති පිළිබඳ විශේෂ හමුවක්
- > රජය නව ආර්ථික සැලසුම් හඳුන්වා දෙයි



Short Summary

කොළඹ මාධ්‍ය හමුවකදී නව ආර්ථික ප්‍රතිපත්ති ප්‍රකාශයට පත් කෙරිණි.



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Result

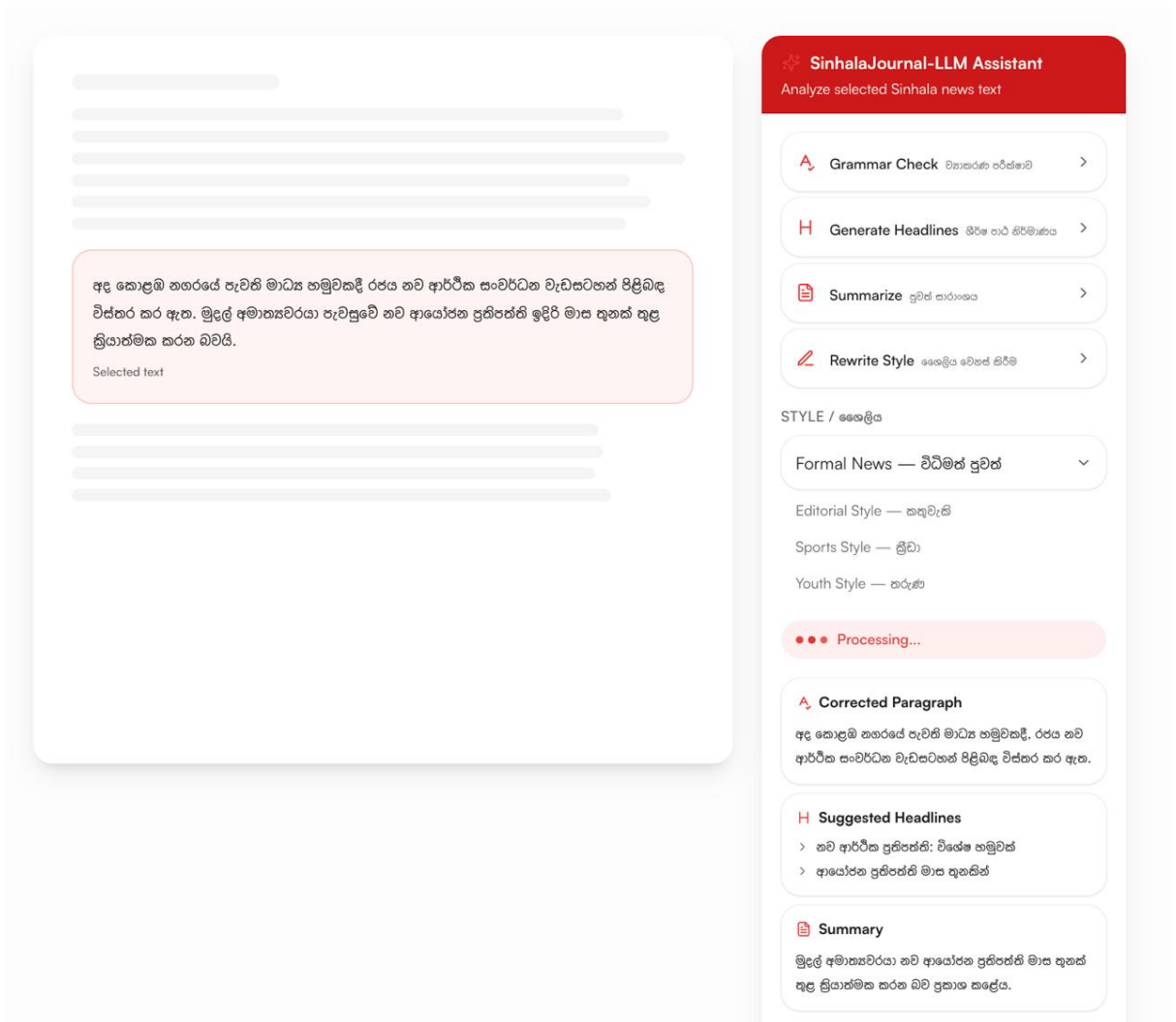


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Editor

- Google Doc



11.2 Risk analysis

Appendix 2: Risk analysis

The development of the Sinhala Style-Adaptive Headline and Image Generator involves several potential risks that may affect system performance and deployment.

1. Technical Risk:

The model may generate inaccurate or misleading headlines, especially in low-resource Sinhala contexts.

Mitigation: Controlled decoding, grammar validation, and human evaluation will be applied.

2. Multimodal Misalignment Risk:

Generated images may not fully align with headline meaning.

Mitigation: A semantic validation step will check alignment before final output.

3. External API Dependency Risk:

The image generation module depends on an external API (e.g., Gemini), which may introduce rate limits or future pricing changes.

Mitigation: Error handling and scalable pricing strategies will be considered.

4. Data Bias Risk:

Training data may contain stylistic or reporting bias.

Mitigation: Dataset balancing and manual review will be conducted.

5. Infrastructure Risk:

GPU resource limitations may affect performance during fine-tuning.

Mitigation: Parameter-Efficient Fine-Tuning (LoRA) reduces computational requirements.

Overall, identified risks are manageable through structured validation, responsible AI practices, and scalable system design.

11.3 Ethical documents

Appendix 3: Ethical documents

Page 2 of 45 - Integrity OverviewSubmission ID: trn:oid::3618:131147272

8% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Small Matches (less than 8 words)

Match Groups

-  **46 Not Cited or Quoted 7%**
Matches with neither in-text citation nor quotation marks
-  **6 Missing Quotations 1%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 4%  Internet sources
- 2%  Publications
- 7%  Submitted works (Student Papers)